Math 3790H: Assignment II

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Problem 1. Let $s_1 = 1$ and $s_{n+1} = \frac{1}{3}(s_n + 1)$ for $n \geq 1$.

(a) Find s_2, s_3 and s_4.

(b) Use induction to show $s_n > 1/2$ for all n. [4]

(c) Show that (s_n) is a decreasing sequence. [3]

(d) Show $\lim s_n$ exists and find $\lim s_n$. [5]

Solution 1.

Problem 2. Let $r \in \mathbb{R}$ with $0 < r < 1$. Suppose (s_n) is a sequence having the property that

$$|s_{n+2} - s_{n+1}| \leq r|s_{n+1} - s_n|$$

for all $n \in \mathbb{N}$.

(a) Show that (s_n) is a Cauchy sequence.

(b) Give an example showing that this can fail for $r = 1$.

Solution 2.

(a) Consider that

$$|s_3 - s_2| \leq r|s_2 - s_1|$$

then, repeatedly chaining the condition given

$$|s_{k+2} - s_{k+1}| \leq r|s_{k+1} - s_k| \leq r^2|s_k - s_{k-1}| \leq \cdots \leq r^{k-1}|s_2 - s_1|.$$

Consider then $m, n \in \mathbb{N}$ with (WLOG) $m > n$. Then

$$|s_m - s_n| = |s_m - s_{m-1} + s_{m-1} - s_{m-2} + \cdots + s_{n+1} - s_n|$$

and by triangle inequality

$$|s_m - s_n| \leq |s_m - s_{m-1}| + |s_{m-1} - s_{m-2}| + \cdots + |s_{n+1} - s_n|.$$

Then we have that $|s_m - s_{m-1}| \leq r^{m-2}|s_2 - s_1|$, $|s_{m-1} - s_{m-2}| \leq r^{m-3}|s_2 - s_1|$, etc. up to $|s_{n+1} - s_n| \leq r^{n-1}|s_2 - s_1|$. Certainly then $|s_m - s_n|$ is bounded above by the sum of all of these bounds,

$$|s_m - s_n| \leq \left[r^{m-2} + r^{m-3} + \cdots + r^{n-1}\right]|s_2 - s_1|.$$

If we factor out an r^{n-1}, and reverse the order so it's more clear,

$$|s_m - s_n| \leq r^{n-1}\left[1 + r + r^2 + \cdots + r^{m-n-2} + r^{m-n-1}\right]|s_2 - s_1|$$

which is a geometric series from 0 to $m - n - 1$. Such a geometric series is, for the same r, bounded above by its infinite form,

$$\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$$

which tells us that

$$|s_m - s_n| \leq \frac{r^{n-1}}{1-r}|s_2 - s_1|.$$

Now the definition of a Cauchy sequence is that for each $\epsilon > 0$ (I still don't quite see the difference between for each and for all in this case, but whatever) there exists an $N \in \mathbb{N}$ such that $m, n > N$ implies that

$$|s_m - s_n| < \epsilon.$$

Certainly if

$$|s_m - s_n| \leq \frac{r^{n-1}}{1-r}|s_2 - s_1|,$$

then N, which is smaller than n, is also suitable for an upper bound as $r^{N-1} > r^{n-1}$ so

$$|s_m - s_n| < \frac{r^{N-1}}{1-r}|s_2 - s_1|$$

by making N arbitrarily large, we can then force $|s_m - s_n|$ arbitrarily small, meaning s_n is Cauchy?

(b) Give an example showing that this can fail for $r = 1$.